



6118 - An Empirical Constraint on the Evolution of An Isolated Low-Mass Galaxy: The Star Formation History of the Dwarf Galaxy Sextans A

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

In the nearby Universe low-mass galaxies are some of the best probes of galaxy formation and evolution, and are often used as benchmarks for theory and cosmological simulations. However, low-mass satellites near a massive host evolve in complex environments and undergo environmental processing that can dominate the shape of their SFHs, making their secular evolution difficult to constrain. In contrast, isolated low-mass galaxies evolve in simpler environments and are less influenced by environmental processing, thereby providing a clearer picture on secular galaxy growth. Yet, to date, we have measured robust SFHs for only four isolated dwarf galaxies across the stellar mass range from 10^5 - 10^8 Msun.

We propose to derive the SFH of the isolated, low-mass, well-studied galaxy Sextans A. We will use archival NIRCcam imaging of the resolved stars in Sextans A that are incredibly deep and well-suited for SFH work. The resulting constraints will allow us to shed new light on the evolution of a

low-mass galaxy outside the influence of a massive host, investigate tentative evidence that there was an early dynamical interactions between Sextans A and the Milky Way (MW), and explore the interplay of low-mass galaxies and reionization.

OBSERVING DESCRIPTION